

Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,16,2);

int GreenState = 0;
int YellowState = 0;
int SupplyState = 0;
void setup() {
  Serial.begin (9600);
  lcd.init();
  lcd.backlight();
  lcd.clear();

  pinMode(9, OUTPUT); //SOLENOID SUPPLY
  pinMode(8, OUTPUT); //PUMP
  pinMode(10, INPUT);
  pinMode(11, INPUT);
  pinMode(12, INPUT);

  lcd.setCursor(0,0);
  lcd.print("WaterTank System");
  delay(4000);
  lcd.clear();

  digitalWrite(9, LOW);
  digitalWrite(8, LOW);
  lcd.setCursor(0,0);
  lcd.print("PUMP: OFF");
  lcd.setCursor(0,1);
  lcd.print("SUPPLY OFF");
}
```

Fig. 3: Arduino code part 1

```
void loop() {
  GreenState = digitalRead(11);
  YellowState = digitalRead(10);
  SupplyState = digitalRead(12);
  if (SupplyState == HIGH ) //SUPPLY IS ON
  {

    if (GreenState == HIGH) //WATER LEVEL BELOW 100%
    {
      digitalWrite(9, HIGH);
      digitalWrite(8, LOW);

      lcd.clear();
      lcd.setCursor(0,0);
      lcd.print("PUMP: OFF");
      lcd.setCursor(0,1);
      lcd.print("SUPPLY ON");
    }
    else
    {
      digitalWrite(9, LOW);
      digitalWrite(8, LOW);

      lcd.clear();
      lcd.print("PUMP: OFF");
      lcd.setCursor(0,1);
      lcd.print("SUPPLY OFF");
    }
  }
}
```

Fig. 4: Arduino code part 2

BILL OF MATERIAL

Components	Quantity
Arduino Nano (MOD1)	1
16x2 LCD display with I2C/IIC interface (MOD2)	1
12V, 2A SMPS	1
LM2596 DC-DC buck converter (manually set to 5V)	1
Solenoid valve (L1)	1
10A, 240V, 2-pole contactor	1
IRF530 n-channel MOSFET (Q1, Q2)	2
BC548 BJT (Q3, Q4, Q5)	3
1N5402 diode (D1, D2)	2
10A10 diode (D3)	1
Green, yellow, and red LED	1 Each
100-ohm resistor (R4, R5, R6)	3
1.5k resistor (R8, R9)	2
2.2k resistor (R1, R2, R3)	3
10k resistor (R7, R10)	2
1x3 JST connector for the water level probe	1
DC jack to connect solenoid valve	1
220V 3-pin connector for water pump	1
AC 220V, 10A 3-pin panel mount plug socket for 220V input	1
10A rated switch	1
5V, one changeover relay	2
MECHANICAL COMPONENTS	
1HP household water pump	1
PCV pipe (cut to different sizes as per diagram)	1 metre (min)
One-way valve	2
Elbow joint	2
Female copper adaptor	4
Steel hose nipple	3

head tank to the desired level using any of the two available sources. The filling of the tank through municipal supply is controlled through a solenoid valve and the water pump is controlled through a contactor.

The decision making is done through a microcontroller in Arduino Nano and electronic devices like BJT, MOSFETs, and diodes. The working of the project can be explained through two cases that utilise different water sources based on their availability.

Case 1. Water level in overhead tank is below 75% and municipal supply is not available. In this case, water

will be filled through the underground tank using water pump till 75% water level is reached.

Case 2. Water level in overhead tank is below 100% and municipal supply is available. In this case, water will be filled through the municipal supply to the tank's full capacity. A solenoid valve will be used to stop filling the tank after the tank is fully filled.

Fig. 1 shows the basic design idea of project while circuit diagram of the automatic water system is shown in Fig. 2. The electronic and mechanical components required to make this project are listed in the Bill of Material table.

Circuit and working

The circuit is built around Arduino Nano (MOD1), 16x2 LCD display